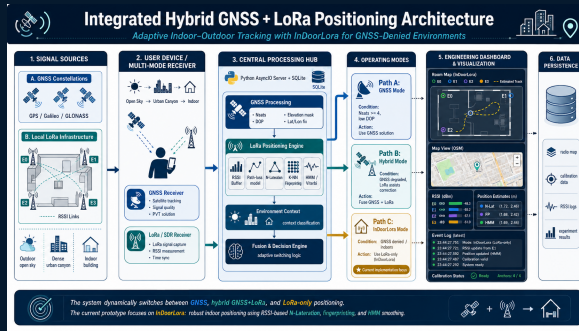


InDoorLora

Hybrid GPS + LoRa Positioning System

From outdoor GPS to indoor LoRa-only tracking



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Supervised by: Prof. Philippe Canalda

Project Storyline

Past

Research architecture

The first step was the hybrid concept: GPS/GNSS for outdoor positioning, LoRa assistance for degraded areas, and LoRa-only mode when GNSS is denied.

Now

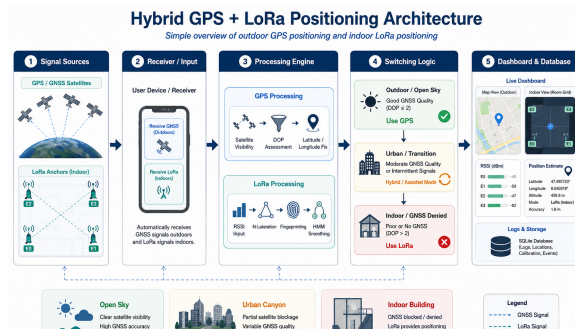
Working prototype

The current system is InDoorLora: a real-time indoor positioning prototype using four anchors, RSSI, Python processing, SQLite, and a live dashboard.

After

Data and ML

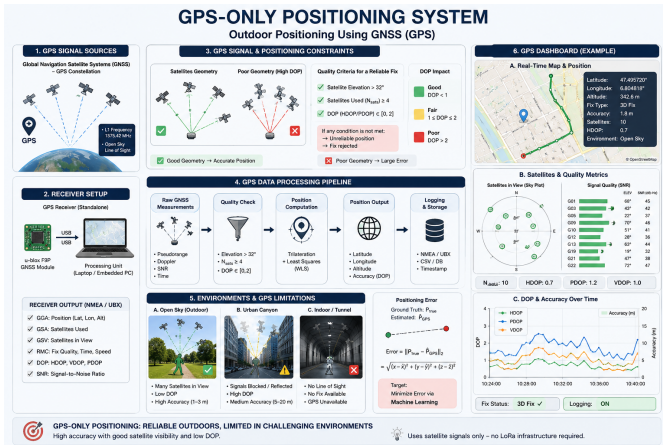
The next step is to collect clean datasets, add ground truth, train correction models, and evaluate accuracy with error curves.



1. Why We Need Hybrid Positioning

GPS works well outdoors, but it becomes weak in urban canyons and usually unavailable inside buildings.

GPS-Only Positioning



Key points

- ▶ GPS/GNSS is accurate in open-sky conditions.
- ▶ Quality depends on satellite visibility, geometry, and DOP.
- ▶ Urban canyons increase errors because of blockage and multipath.
- ▶ Indoors, GPS is usually unavailable.

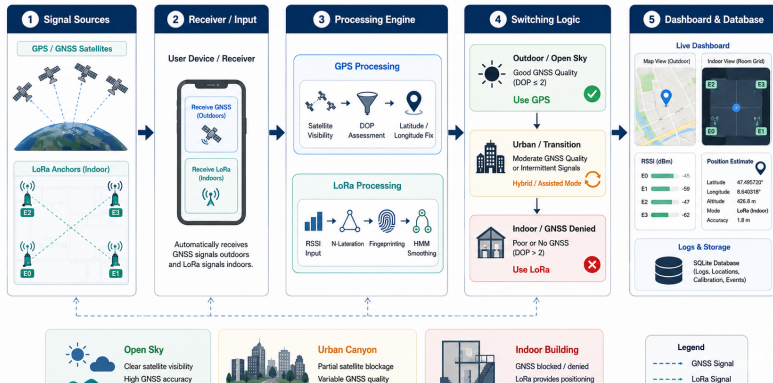
Open sky

Indoor limit

Simple Hybrid GPS + LoRa Overview

Hybrid GPS + LoRa Positioning Architecture

Simple overview of outdoor GPS positioning and indoor LoRa positioning



System idea

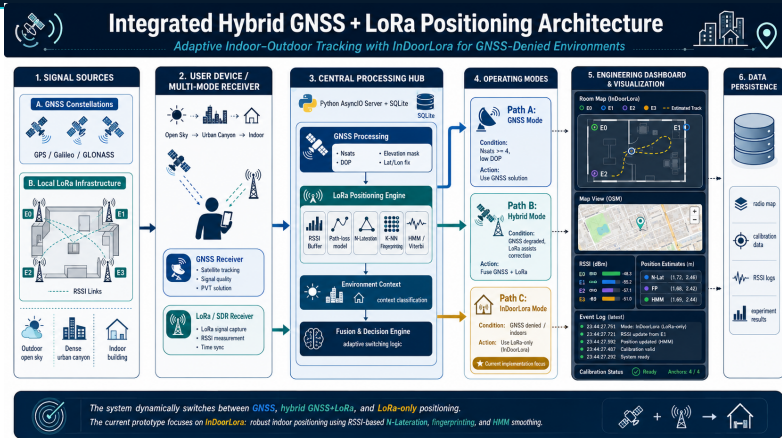
- ▶ Use GPS when outdoor quality is good.
- ▶ Use hybrid mode when GPS is degraded.
- ▶ Use LoRa when GNSS is denied indoors.
- ▶ Show both outdoor and indoor views in the dashboard.

GPS

LoRa

Switching

Full Hybrid Architecture



The complete vision: GNSS sources, local LoRa infrastructure, receiver, processing hub, mode switching, dashboard, and persistent data.

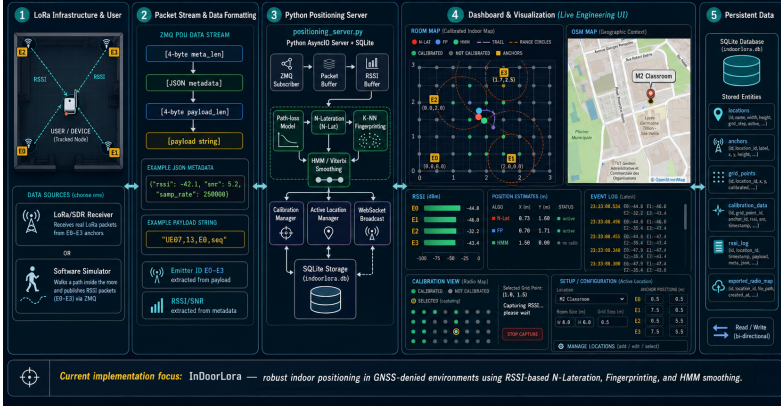
2. What We Built Now

The current implementation is the indoor LoRa layer: InDoorLora. It focuses on real-time RSSI-based positioning in GNSS-denied environments.

InDoorLora Architecture

InDoorLora: LoRa Indoor Positioning System Architecture

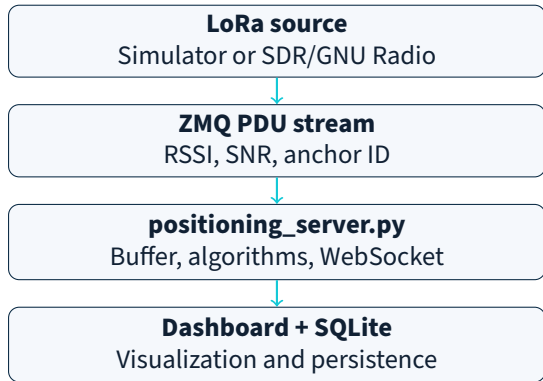
Real-time indoor positioning using 4 LoRa anchors, Python processing, SQLite persistence, and a live engineering dashboard



Current implementation focus: InDoorLora — robust indoor positioning in GNSS-denied environments using RSSI-based N-Lateration, Fingerpringting, and HMM smoothing.

Current implementation: four anchors, ZMQ packet stream, Python positioning server, SQLite storage, and live dashboard visualization.

Current Software Pipeline



Implemented algorithms

- ▶ **N-Lateration:** converts RSSI to distance and estimates position geometrically.
- ▶ **Fingerprinting:** compares the live RSSI vector with the calibrated radio map.
- ▶ **HMM / Viterbi:** smooths the trajectory and reduces unrealistic jumps.

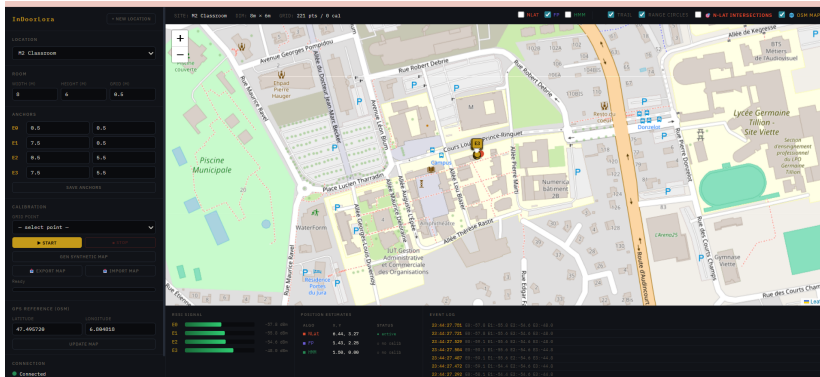
ZMQ

WebSocket

SQLite

HMM

Dashboard: OSM View



Live system

- ▶ Active site: M2 Classroom.
- ▶ Room dimensions: 8 m by 6 m.
- ▶ OSM view gives geographic context.
- ▶ RSSI and estimates are shown in real time.
- ▶ This view connects indoor data with the outdoor map.

Location and Anchor Management

InDoorLora + NEW LOCATION

LOCATION

MyRoom ▼

- select -

M2 Classroom

MyRoom

ANCHORS

E0	0	0
E1	2	0
E2	0	2
E3	1.7	2.5

SAVE ANCHORS

NEW LOCATION

NAME

e.g. Lab B204

WIDTH (M) HEIGHT (M) GRID STEP (M)

8.0 6.0 0.5

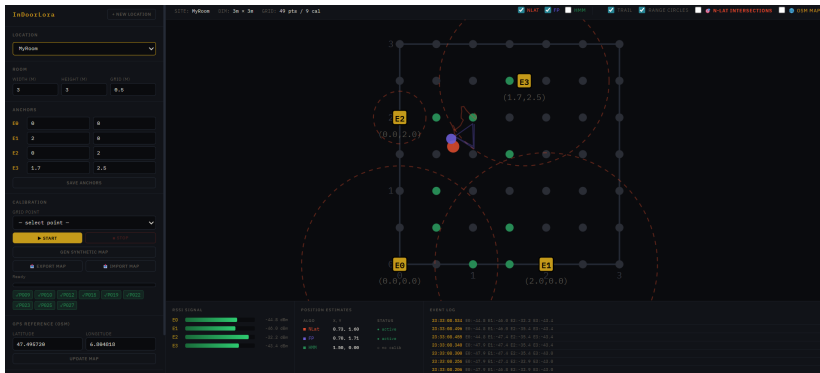
ANCHOR POSITIONS

E0	0.5	0.5
E1	7.5	0.5
E2	0.5	5.5
E3	7.5	5.5

CANCEL CREATE LOCATION

The dashboard supports multiple rooms. Each location has its own dimensions, grid step, anchors, and calibration map.

Live Indoor Tracking in MyRoom



Current prototype

- ▶ Four local anchors: E0, E1, E2, E3.
- ▶ Green points represent calibrated grid nodes.
- ▶ Range circles visualize N-Lateration constraints.
- ▶ RSSI bars show signal strength from each anchor.
- ▶ N-Lat, FP, and HMM are displayed together.

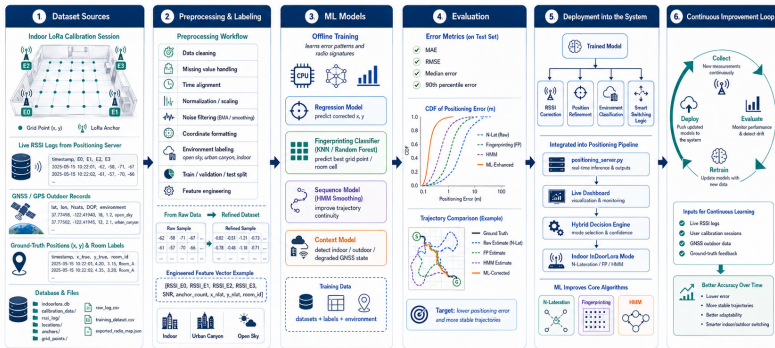
3. How Datasets and ML Improve It

The next step is to turn calibration logs and ground-truth trajectories into models that reduce positioning error.

Datasets and Machine Learning Loop

Using Datasets and Machine Learning to Improve Positioning Accuracy

How InDoorLora and hybrid positioning data are transformed into ML models that improve the system



Datasets + calibration + machine learning = smarter positioning, lower error, better indoor/outdoor adaptation.

Datasets + calibration + ML create smarter positioning: lower error, more stable trajectories, and better indoor/outdoor adaptation.

Dataset Plan

Training data

- ▶ Timestamp
- ▶ Room ID and grid point ID
- ▶ True position: x_{true}, y_{true}
- ▶ RSSI from E0, E1, E2, E3
- ▶ SNR when available
- ▶ Environment label: indoor, outdoor, transition

Calibration data

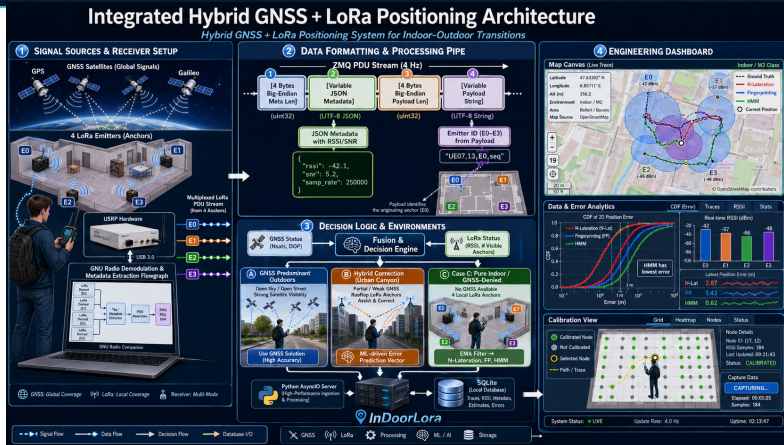
RSSI logs

Evaluation data

- ▶ Separate from calibration data.
- ▶ Uses known ground-truth paths.
- ▶ Compares N-Lat, FP, HMM, and ML-corrected output.
- ▶ Produces error curves and trajectory plots.

$$Error = \sqrt{(x - \hat{x})^2 + (y - \hat{y})^2}$$

Full Detailed System View



Recap of the complete vision: GNSS, LoRa, SDR/USRP, PDU formatting, decision logic, dashboard, calibration, and performance analytics.

Past, Now, After

Past

- ▶ Hybrid GNSS + LoRa idea.
- ▶ GNSS quality criteria.
- ▶ Case A, B, and C logic.
- ▶ Architecture diagrams.

Concept



Now

- ▶ InDoorLora prototype.
- ▶ Python server and dashboard.
- ▶ SQLite persistence.
- ▶ N-Lat, FP, HMM.

Prototype



After

- ▶ Clean datasets.
- ▶ Ground-truth trajectories.
- ▶ ML correction models.
- ▶ Accuracy evaluation.

Measured performance

Current Limitations and Next Steps

Working now

- ▶ Live dashboard.
- ▶ Multi-room configuration.
- ▶ Anchor coordinate management.
- ▶ RSSI packet pipeline.
- ▶ Three algorithms.
- ▶ OSM visualization.

Next work

- ▶ Clean the database and logs.
- ▶ Complete calibration data.
- ▶ Record ground-truth paths.
- ▶ Validate with real LoRa/SDR hardware.
- ▶ Add ML correction.
- ▶ Report numerical accuracy.

Conclusion

InDoorLora is the working indoor LoRa layer of the broader Hybrid GPS + LoRa project.

It already demonstrates real-time RSSI processing, localization algorithms, dashboard visualization, calibration, and persistent storage.

The next challenge is data quality: calibration, ground truth, ML correction, and measured accuracy.

Concept → Prototype → Evaluation